

❏ Full Theory: IDS Architecture, Agents, Sensors, DoS/DDoS

❏ 1. Symptoms of Attacks

Attack symptoms are behavioral or system-level **indicators** that something is wrong. Some common symptoms:

Symptom	Description
❏ Slow network	Possible DDoS or exfiltration
❏ Unusual traffic	Large data flows, odd protocols
❏ Unauthorized access	Logs showing unknown logins
❏ Sudden disk usage	Possible ransomware
❏ Application crashes	Buffer overflows, exploit attempts
❏ Reboots or hangs	Rootkits or low-level malware
❏ Unexpected inbound connections	Could indicate backdoor or RAT

❏ 2. Tiered (Tired) IDS Architecture

(Tired is likely a typo for Tiered Architecture)

A **tiered IDS architecture** divides functions into logical levels:

❏ 1. Data Collection Layer (Sensors)

- Collects packets, logs, system calls
- Deployed on hosts or networks

❏ 2. Analysis Layer (Agents)

- Filters and interprets collected data
- Applies detection rules or behavior models

❏ 3. Management Layer (IDS Manager)

- Centralized control and coordination
- Alerts, policies, correlation, reporting

This modularity allows scalability, load distribution, and easier maintenance.

❏ 3. Sensors – Network vs Host-Based

Sensor Type	Description
Network-Based Sensor (NIDS)	Captures packets, inspects traffic
Host-Based Sensor (HIDS)	Monitors logs, processes, file changes
Wireless Sensor	Monitors wireless channels
Application Sensor	Tracks application-layer anomalies (e.g., web requests)

Sensors are **deployed at strategic points**: gateways, DMZs, endpoints.

⌘ 4. Denial of Service (DoS)

A **DoS attack** aims to **make a service unavailable** to legitimate users. Types include:

- **Flood attacks** (ICMP, UDP, SYN flood)
- **Logic attacks** (Ping of death, Slowloris)
- **Application-layer attacks** (HTTP GET flood)

⌘ DDoS (Distributed DoS):

Same idea but launched from **multiple sources** (botnets), making it harder to block.

⌘ 5. DoS & DDoS Mitigation Techniques

Mitigation	Description
⌘ Rate Limiting	Restricts number of requests from source
⌘ Geo/IP Filtering	Blocks traffic from malicious regions
⌘ Firewalls & IPS	Drop suspicious packets
⌘ Reverse Proxies/CDNs	Absorb/buffer high traffic loads
⌘ Scrubbing Centers	Redirect traffic to third-party filters
⌘ Blackhole Routing	Drops all packets for target under attack (last resort)
⌘ Anomaly Detection IDS	Detects spikes in traffic volume or packet anomalies

❏ 6. Sensor Deployment Strategy

Effective sensor deployment should ensure **visibility and coverage**:

- **Perimeter** – Catch inbound/outbound traffic
- **Core switches** – Internal lateral movement
- **Critical hosts** – File/system/process monitoring
- **DMZ** – Monitor exposed servers
- **Wireless APs** – Prevent rogue access points

Avoid bottlenecks, mirror ports, and use **SPAN/TAP** methods for NIDS.

❏ 7. Agents in IDS

Agents are software components responsible for processing, analyzing, or reacting to collected data. Types:

Agent Type	Role
Sensor Agents	Collect data from host or network
Analyzer Agents	Apply rules, detect patterns
Alert Agents	Notify central manager
Response Agents	Block/quarantine threats
Communication Agents	Relay data to manager

❏ 8. Functions of IDS Agents

Agents perform key IDS functions:

- Data collection & normalization
- Packet filtering
- Log parsing (Syslog, Windows Events)
- Rule matching (signatures or anomalies)
- Alert generation and forwarding
- Local response (quarantine, kill process)

☒ *Distributed agents reduce load on central system and increase detection granularity.*

☒ 9. IDS Manager

The **IDS Manager** is the **brain of the system**.

It:

- Receives alerts from agents
- Correlates and prioritizes events
- Stores logs
- Visualizes attack patterns
- Sends alerts (email, dashboard)
- Applies or updates policies

Tools like **Snort + Snorby**, **OSSEC**, or **AlienVault** use a central manager to orchestrate detection.